Bubble Sort Time Complexity

```
#include <iostream>
using namespace std;

void asc_bubble_sort(int arr[], int size) {
      for(int i=0;i<size-1;i++){
        for(int j=0;j<size-i-1;j++){
           if(arr[j]>arr[j+1])
             swap(arr[j],arr[j+1]);
        }
      }
}
void show(int arr[], int size) {
   for (int i = 0; i < size; i++) {
      cout << arr[i] << " ";
   }
   cout << endl;
}

int main() {

   int arr[5]={5,12,0,1,8};
   asc_bubble_sort(arr, 5);
   show(arr, 5);

   return 0;
}---> calculate  the time complexity of that bubble sort using comparison calculation with iterative simulation
```

We will analyze the time complexity of **Bubble Sort** using a structured breakdown of steps, as you prefer.

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1. Understanding the Algorithm

Bubble Sort is a simple sorting algorithm that works by repeatedly swapping adjacent elements if they are in the wrong order. It consists of multiple passes through the array.

For an array of size N:

- The **outer loop** runs $N - 1$ times.
- The **inner loop** compares adjacent elements and swaps them if necessary.
- The number of comparisons decreases in each pass.

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2. Number of Comparisons Calculation

We simulate the comparisons step by step for an array of size $N = 5$.

General Comparison Formula

- In the first pass ($i = 0$), we compare the first $N - 1$ elements $\rightarrow N - 1$ comparisons.
- In the second pass ($i = 1$), we compare the first $N - 2$ elements $\rightarrow N - 2$ comparisons.
- In the third pass ($i = 2$), we compare the first $N - 3$ elements $\rightarrow N - 3$ comparisons.
- In the fourth pass ($i = 3$), we compare the first $N - 4$ elements $\rightarrow N - 4$ comparisons.

So, the **total number of comparisons** for an array of size N is:

$$(N-1) + (N-2) + (N-3) + \cdots + 2 + 1$$

This is the sum of the first $N - 1$ natural numbers:

$$\sum_{i=1}^{N-1} i = \frac{(N-1)N}{2}$$

For $N = 5$:

$$\frac{(5-1) \times 5}{2} = \frac{4 \times 5}{2} = 10 \text{ comparisons}$$

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3. Big-O Notation Analysis

The sum formula $\frac{N(N-1)}{2}$ simplifies to $O(N^2)$ in asymptotic notation, ignoring constants.

Thus, the worst-case time complexity of **Bubble Sort** is:

$$O(N^2)$$

Summary

- **Total comparisons for $N = 5$: 10**
- **General formula**: $\frac{N(N-1)}{2}$
- **Big-O Complexity**: $O(N^2)$

Would you like a breakdown of swaps as well?